FormSet and Parallel Derivation: a synthesis

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December 14, 2025

Abstract

This paper analyzes the FormSet operation as it has been defined and used in the literature specifically in the domain of conjunction structures. It finds that, though its use as a structure building operation must either contradict its definition and justification as a general-purpose preparatory operation or lead to unwanted empirical predictions. It goes on to show that, if we redefine MERGE as applying not to syntactic objects in a workspace but to n-ary sets of syntactic objects in a workspace, that we can retain the justification of FormSet without sacrificing empirical validity. Furthermore, it presents novel empirical arguments in favour of this redefined MERGE—that it correctly predicts the ambiguities raised by conjunction structure, that it correctly predicts a cross-linguistic gap in the plural anaphor system, and that it gives some insight into the system of number "features."

I

In recent theoretical work, Chomsky (2021, 2024) proposed FormSet, a new structure building operation (SBO) which forms arbitrarily large sets of mental objects[1]. These sets, it is proposed, are treated by the language faculty in much the same manner as individual objects. That is, n-member sets can be the input to syntactic merge, and thus can explain heretofore unexplained grammatical phenomena, most pertinently, coordination. So, for instance, the conjoined subject in (1) is analysed as a set that contains the conjuncts and the coordinator and is merged directly with the predicate.

(1) Geri, her mother and the man who designed the house arrived.

(2) $\{\{\text{and}, \text{Geri}, \text{her mother}, \text{the man who designed the house}\}, \{past, \{\ldots, \text{arrive}\}\}\}$

This simple proposal raises two potential objections—one theoretical, the other empirical. The theoretical objection is one of minimalism—SBOs other than Merge require more than merely empirical justification. That is, it is not enough to assert that a SBO is sufficient to account for this or that phenomenon, one must show either that it is necessary, or, as Chomsky does in this case, that our theory already assumes this SBO. Indeed, FormSet is indispensable if our theory is to include things like workspaces or the lexicon.

The empirical problem arises when we bring in some standard definitions in syntactic theory...

*I am indebted to members of Linguistics communities at the University of Toronto, University of Chicago, and MIT as well as Erik Zyman for comments previous versions of this paper.

[1] Initially, he proposed it as FormSequence, which constructed ordered sets.

(3) Merge(x, y) = $\{x, y\}$ where
x and y are syntactic objects, and
x and y are separate (external merge) or x is a term of y (internal merge).

(4) x is a syntactic object iff
x is a lexical item or
x is a set of syntactic objects

(5) x is a term of y iff
x is a member of y or
there is some z such that z is a member of y and x is a term of z.

...and a standard coordinate structure constraint violation.

(6) * Which movie did David praise *Jaws* and critique t_{wh}?

Under a naive FormSet analysis, (6) is derived by internal merge from (7).

(7) $\{\mathrm{C}_{wh}, \{\mathrm{David}, \{\ldots\{\mathrm{and}$, praise *Jaws*, critique which movie$\}\ldots\}\}\}$

This would be a completely valid instance of internal merge under this analysis, and therefore would amount to falsification of the analysis.

At this point, one may object that the CSC, at least as a narrowly syntactic phenomenon, has been soundly disproved—that myriad counterexamples have been presented, and that, this being the case, a theory of coordination that gives a narrowly syntactic account of the CSC is flawed. To this I give a response that echoes that given by Milway (2022) in response to Truswell's (2011) purported refutation of the Adjunct Condition: Suppose we concede that certain apparent cases of coordination are not subject to the CSC. We would then need to explain or describe what makes the cases in question exceptional relative to coordination structures, and normal relative to ordinary hierarchical structures. We would not be off the hook, though, as we would be left with a class of structures that *are* subject to the CSC that require explanation—an explanation that we are presently attempting.

There are, however, alternative proposals for FormSet, described by Fong and Oishi (2023) and Marcolli, Chomsky, and Berwick (2025) respectively, which are more purpose-built to avoid such an issue. Fong and Oishi (2023), despite identifying their version of FormSet with the version of the operation described above, deviate from Chomsky by requiring that the operation is constrained by what they call "parallelism". Marcolli, Chomsky, and Berwick (2025), on the other hand, are more explicit in how their version deviates from Chomsky's (2021; 2024)—the FormSet they propose is different from the operation that constructs workspaces—and justify this deviation, writing

> The reason why it is reasonable to keep these two operations distinct lies in the fact that the workspace is just a "transitory" structure that defines the locus where the steps of a derivation that build the final resulting sentence structure are taking place. On the other hand, the grouping together of components via FormSet creates a structure that remains in the ultimate form of the resulting sentence[.] (146)

The details of these latter proposals are immaterial, because even if we take for granted their descriptive adequacy, they still fail under minimalist scrutiny, as they represent a distinct narrow-syntactic operation from Merge. Indeed, it is hard to dispute the initial justification for FormSet as a general cognitive operation that precedes Merge both logically and in terms of evolutionary history

—our minds simply must be able to form sets of mental objects. Once FormSet is complicated and specialized by a theorist, however, it loses that justification.

In an alternate theory, Milway (2022) proposes a mechanism of Parallel Derivation to account for adjunction, though it can be extended to coordination. According to this theory, (1) consists of a set of syntactic objects in (8) rather than merely containing a set as a subject as in (2).

(8) $$\left\{\begin{array}{l}\{\text{Geri}, \{past, \{\text{Geri}, \text{arrive}\}\}\}, \\ \{\text{her mother}, \{past, \{\text{her mother}, \text{arrive}\}\}\}, \\ \{\text{the man who designed the house}, \{past, \{\text{the man who designed the house}, \text{arrive}\}\}\}\end{array}\right\}$$

Milway (2022) shows how this theory can account for a number of empirical facts of adjunction including those that are mirrored in the domain of coordination, but achieves this mainly by introducing the higher-order function `map`, which applies a function to the members of a list and results in another list. If we are able to attain these empirical properties without adding additional operations, though, we ought to.

We can retain FormSet and Parallel Derivation as hypotheses, however, if we introduce and slightly alter another hypothesis from Chomsky (2020), namely MERGE, or the operation of merge on workspaces. MERGE can be considered a version of merge that operates not on free-floating syntactic objects, but on workspaces and the objects they contain. Furthermore, MERGE is constrained by certain third-factor considerations. In particular the general computational principle of Resource Restriction, when applied to language, "guarantees that Merge yields the fewest possible new terms that are accessible to further operations[.]" (Chomsky et al. 2023, p. 29) Thus, MERGE can be characterized as in (9).

(9) MERGE(WS, X, Y) $\rightarrow$ WS′ where
  a. X $\in$ WS,
  b. Y $\in$ WS or Y is a term of X,
  c. $\{$X,Y$\}$ $\in$ WS′, and
  d. X, Y $\notin$ WS′

It is important to note that, though MERGE is sometimes construed as a replacement for or a refinement of merge, it actually contains merge in the sense that it combines two distinct syntactic objects—X and Y in (9)—to form a new object containing them—$\{$X,Y$\}$ in (9)—and thus maintains the distinction between the first-factor merge and the third-factor additions that allow merge to operate. Thus we are free to reformulate MERGE under two constraints. The first constraint is that it must conform to other third-factor considerations, such as Resource Restriction. The second is that it must contain first-factor merge at its core—it ought to create binary sets and ought not create ordered pairs or impose labels or the like. With those caveats, we can proceed to formulate MERGE such that it is compatible with FormSet and hopefully can account for unstructured expressions like coordination.

Before continuing though, it is necessary for us to define a notational convention. Up to now we have been representing all sets, regardless of which operation created them. Since we will be distinguishing between FormSet sets and merge sets, that distinction ought to be represented in our notation. To that end, from this point on, curly brackets $\{\cdot\}$ will indicate FormSet sets, while square brackets $[\cdot]$ will indicate merge sets (*i.e.*, complex syntactic objects).

II

If we start with the form of MERGE in (10) and assume that WS and WS′ are workspaces (*i.e.*, FormSet sets) we can ask what sorts of things X and Y are.

(10) MERGE(WS, X, Y) $\rightarrow$ WS′

We can follow Chomsky (2021, p. 30)[2] in his proposal that XPs are the "limiting case" of those FormSet sets when MERGE applies. That is, we can assume that X and Y in (10) are singleton sets of SOs—in most cases {XP} and {YP}, respectively—and generalize from there. In this case, the resulting WS′ must contain the new SO [XP, YP] as in (11).

(11) MERGE(WS, {XP}, {YP}) $\rightarrow$ {[XP, YP], . . . }

Note that we differ from Chomsky (2021) who assumes that FormSet sets persist throughout the derivation. Instead, FormSet is applied SOs in the workspace in preparation for MERGE.

There is a simple way to generalize this to cases where X and Y are n-ary sets—applying Merge to the Cartesian product X×Y as in (12).

(12) MERGE(WS, X, Y) $\rightarrow \{\text{merge}(x, y) | \langle x, y\rangle \in \text{X} \times \text{Y}\} \cup (\text{WS} - \{\text{X,Y}\})$

To take a concrete case, the coordination in (1) would still be achieved by applying MERGE to the FormSet set {*Geri, her mother, the man who designed the house*} and the singleton FormSet set {*arrive*}. The result, however, is not a syntactic object that contains FormSet sets, but a set of syntactic objects as shown in (13).

(13) MERGE(WS, {*Geri, her mother, the man who designed the house*}, {*arrive*})

$$\rightarrow \left\{\begin{array}{l} [\textit{Geri, arrive}], \\ [\textit{her mother, arrive}], \\ [\textit{the man who designed the house, arrive}], \\ \text{pst, C}, \ldots \end{array}\right\}$$

This, then, would be the underlying (intermediate) structure for a sentence with coordination like (1) under an analysis which reformulates MERGE in a way compatible with a version of FormSet that meets minimalist scrutiny.

III

Though this reformulated MERGE is theoretically sound, its empirical viability is not immediately obvious. The first hurdle to clear is that of externalization. The continuation of the derivation that includes (13) would result is a set of complete syntactic objects which would need to be linearized to (1). If we take for granted that our SM system can handle the linearization of a single SO, and that linearizing an n-ary set is a matter of somewhat arbitrary choice[3] then we are left with the ordered set of strings in (14) which must be mapped to (1).

[2] Chomsky (2021) is writing about FormSequence, the forerunner of FormSet.

[3] There are certainly some restrictions on possible linear orders of conjunctions—for instance, *her mother* cannot precede *Geri* in (1) and maintain co-reference between *her* and *Geri*, or see the work of Benjamin Bruening (*e.g.*, 2014) for other examples. For now, though, we can set these aside as we construct a fuller theory.

(14) ⟨Geri arrived, her mother arrived, the man who designed the house arrived⟩

If we assume that repeated material—*i.e.*, *arrived*—is treated as copies of each other, perhaps due to FormCopy, then we would expect redundant copies to be deleted. Since none of the copies c-command the others, however, we might expect a free choice between which copies to be deleted—that (15) and (16) would also be viable pronunciations.

(1) Geri, her mother and the man who designed the house arrived.

(15) * Geri arrived, her mother and the man who designed the house.

(16) * Geri, her mother arrived and the man who designed the house.

When we consider the linearization process that would yield (15) and (16), however, we can see that they are suboptimal. Taking (14) as a stage in the linearization process, we can see that linear precedence relations between all of the morphemes have been set—within the individual clauses, by the LCA or whatever similar process derive linear order on unordered trees, and between the clauses, some arbitrary choice. So, for instance, the noun *house* precedes the verb *arrived*.

In both (15) and (16), though *arrived* precedes *house*. An optimal linearization process would likely not involve such a reordering of the morphemes unless there was some compelling reason to do so. Since there is no such reason in these cases, it is reasonable to conclude that (15) and (16) would be ruled out as suboptimal linearizations.

Next we can consider how this conception of MERGE fares with respect to the CSC. To begin, let us take the would-be derivation of (6) starting at the point prior to the illicit *wh*-movement.

(17) $\left\{\begin{array}{l}[\mathrm{C}_{Wh}, [\textsc{past}, [\text{David praise } \textit{Jaws}]]], \\ [\mathrm{C}_{Wh}, [\textsc{past}, [\text{David critique which movie}]]]\end{array}\right\}$

At this point in the derivation, there is a legitimate *wh*-movement to be made, internally merging *which movie* with the SO that contains it resulting in the Workspace (18).

(18) $\left\{\begin{array}{l}[\mathrm{C}_{Wh}, [\textsc{past}, [\text{David praise } \textit{Jaws}]]], \\ [\text{Which movie}, [\mathrm{C}_{Wh}, [\textsc{past}, [\text{David critique which movie}]]]]\end{array}\right\}$

This would leave the C_{wh} in the first member of (18) unsatisfied and the derivation would crash. We could conceivably try to rescue the first member, say by replacing C_{wh} with its polar or declarative version—C_{Q} or $\mathrm{C}_{\emptyset}$ respectively—but this would break the parallelism required to reduce the two SOs to a single freestanding clause containing coordinated expression and result in either two separate sentences or a sentence consiting of two coordinated clauses. Instead we would get something like (19) or (20).

(19) Did David praise *Jaws*? Which movie did he critique?

(20) David praised *Jaws*. Which movie did he critique?

We can also see how our improved conception of MERGE accounts for ATB movement such as in (21) again by starting at the point prior to *wh*-movement, shown in (22)

(21) Which movie did David praise and critique?

(22) $\left\{\begin{array}{l}[\mathrm{C}_{Wh}, [\textsc{past}, [\text{David praise which movie}]]], \\ [\mathrm{C}_{Wh}, [\textsc{past}, [\text{David critique which movie}]]]\end{array}\right\}$

Two options for the next step spring to mind. The first is that each individual *wh*-phrase undergoes internal merge separately. The second is that a single instance of the *wh*-phrase is merged with both by the instance of MERGE in (23)

(23) MERGE(WS, $\{$*Which movie*$\}, \{[\text{C}_{wh}, \ldots], [\text{C}_{wh}, \ldots]\})$

Since these are distinct derivations, though, they to map to distinct interpretations. The first, in which the there are two independent instances of *wh*-movement, we might expect to mean that David praised one movie and critiqued another. The second, in which there is a single *wh*-movement step, we might expect to mean that David praised and critiqued a single movie.

IV

There are additional facts of coordination which may be explained with the proposed version of MERGE but not with the older version. First, consider the systematic ambiguity in sentences like (24) which have multiple instances of coordination.

(24) Tom, Dick and Harry danced and sang all night (Chomsky 2024)

According to Chomsky (2024) "[t]here is no set of conjuncts that expresses the meaning of [(24)]." That is, it does not mean that each of Tom, Dick and Harry both danced and sang but is compatible with, for example, a situation where Tom and Dick danced but only Harry sang. From this, Chomsky concludes that (24) expresses a single thought and has a single SEM.

There are, however, rather sharp boundaries to the ambiguity in question here. To give a concrete example of the boundaries, (24) cannot mean that Tom danced, Dick sang, and Harry did neither, nor can it mean that the three men all sang, but none danced. To generalize, it seems that if $x_1, \ldots, x_n, y_1, \ldots, y_m, z_1, \ldots$ are conjoined constituents in expression S, then they each must contribute significantly to the interpretation of S. While this may seem like an obvious constraint on interpretation, it cannot be taken for granted that a given theory is compatible with it.

Take, for instance the analysis of conjunction put forth by Chomsky (2021, 2024) and developed by Fong and Oishi (2023) and Marcolli, Chomsky, and Berwick (2025). Under this theory (24) would be one SO that contains n-ary sets as constituents as in (25)

(25) $[\text{C}_\emptyset, [\{$*Tom, Dick, Harry*$\}, [\ldots, [\{$*sing, dance*$\}]]]]$

Broadly considered, the semantic representation of this would resemble the Cartesian product of the two n-ary sets given in (26)—holding the rest of the structure constant.

(26) a. $\{P(x) | \langle x, P\rangle \in \{t, d, h\} \times \{\textbf{sang}, \textbf{danced}\}\}$

b. $\{\textbf{sang}(t), \textbf{danced}(t), \textbf{sang}(d), \textbf{danced}(d), \textbf{sang}(h), \textbf{danced}(h)\}$

As a starting point, we could hypothesize that (24) is true iff a (non-empty) subset of (26) is true. This would, of course, be insufficient, as the very interpretations that we need to rule out are such subsets. We would need to further restrict those subsets as in (27)

(27) M is a possible meaning of (24) iff

$$\begin{cases} M \subseteq \{P(x) | \langle x, P\rangle \in \{t, d, h\} \times \{\textbf{sang}, \textbf{danced}\}\} \\ \& M \neq \emptyset \\ \& \forall x \in \{t, d, h\}. \exists P \in \{\textbf{sang}, \textbf{danced}\}. P(x) \in M & \text{(every subject has an predicate)} \\ \& \forall P \in \{\textbf{sang}, \textbf{danced}\}. \exists x \in \{t, d, h\}. P(x) \in M & \text{(every predicate has a subject)} \end{cases}$$

This, of course, is specific to cases with conjoined subjects and conjoined predicates, and does not obviously generalize to all cases of multiply-coordinated expressions. So, while this theory of coordination can be made consistent with the facts at issue, it does not do so naturally.

The version of MERGE in (12), on the other hand, would not need such explicit constraints to account for the facts in question. Indeed, when we consider how the illicit SEMs would be represented under this theory, shown in (28), we see that they are ruled out because they involve illicit syntactic structures, marked with asterisks.

(28) a. $$\left.\begin{cases} [\text{Tom}, [pst, [\text{Tom}, [v, \text{dance}]]]], \\ [\text{Dick}, [pst, [\text{Dick}, [v, \text{sing}]]]], \\ *[\text{Harry}, [pst, [\text{Harry}, v]]] \end{cases}\right\} \approx \text{Tom danced, Dick sang, and Harry did neither}$$

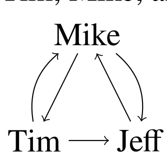
b. $$\left.\begin{cases} [\text{Tom}, [pst, [\text{Tom}, [v, \text{sing}]]]], \\ [\text{Dick}, [pst, [\text{Dick}, [v, \text{sing}]]]], \\ [\text{Harry}, [pst, [\text{Harry}, [v, \text{sing}]]]], \\ *[pst, [v, \text{dance}]] \end{cases}\right\} \approx \text{Tom, Dick and Harry sang, no one danced.}$$

Nor must one put forth elaborate derivational arguments to rule out such structure, as they are ruled out by the theta criterion, with *Tom* lacking a theta-role and *v* associated with *dance* failing to assign its theta-role in the respective structures. In short, the novel theory of coordination naturally accounts for the facts in question.

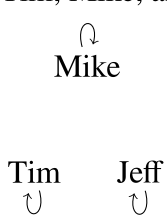
V

Another point of empirical distinction between the two theories of coordination comes from what appears to be a universal gap in the system of plural anaphors that, to my knowledge, has not been remarked upon. Taking English as an example, we see that there is a reciprocal anaphor, *each other*, with a characteristic interpretation represented graphically in (29), and a plural reflexive, *themselves*, with its interpretation represented in (30), but no general plural anaphor, **eachselves*, representing the union of the two others as shown in (31).

(29) a. Tim, Mike, and Jeff like each other.

b. 

(30) a. Tim, Mike, and Jeff like themselves.

b. 

(31) a. * Tim, Mike, and Jeff like eachselves.

b.
$$\begin{tikzcd}
& \text{Mike} \arrow[loop above] \arrow[dl] \arrow[dr, bend left] & \\
\text{Tim} \arrow[ur, bend left] \arrow[rr] \arrow[loop below] & & \text{Jeff} \arrow[ul] \arrow[ll, bend left] \arrow[loop below]
\end{tikzcd}$$

Furthermore, an informal survey of linguists/native speakers[4] shows that this gap exists in Western Armenian, Cantonese, Dutch, Laurentian French , Metropolitan French, Hungarian, Italian, Japanese, Korean, Southern Kurdish, Mandarin, Persian, Russian, Mexican Spanish, and Tagalog. A more nuanced description of the facts is that two broad patterns emerged: One type of language—*e.g.*, English —has distinct lexemes/constructions for reflexives and reciprocals, while the other type—*e.g.*, French —has a single anaphor that is ambiguous between reflexive and reciprocal, with strategies for disambiguating between the two meanings. In all cases, the meaning represented in (31) was expressed with a conjoined anaphor like "themselves and each other".

This gap would be unexpected were we to assume, following Chomsky (2024), that coordinated phrases are represented as n-ary sets since the meaning of the non-existent anaphor **eachselves* is essentially the Cartesian product—*i.e.*, perhaps the most natural way to combine a pair of sets—while the meanings actual anaphors are the constrained subsets of the Cartesian product. If we assume the theory of coordination suggested here, though, it becomes less surprising. Consider what would be the structures of (29) and (30) in (32) and (33), respectively.

(32)
$$\left\{\begin{array}{c}
[\text{Tim}, [pres, [\text{Tim}, [v, [\text{like}, \text{Mike}]]]]], \\
[\text{Tim}, [pres, [\text{Tim}, [v, [\text{like}, \text{Jeff}]]]]], \\
[\text{Mike}, [pres, [\text{Mike}, [v, [\text{like}, \text{Tim}]]]]], \\
[\text{Mike}, [pres, [\text{Mike}, [v, [\text{like}, \text{Jeff}]]]]], \\
[\text{Jeff}, [pres, [\text{Jeff}, [v, [\text{like}, \text{Tim}]]]]], \\
[\text{Jeff}, [pres, [\text{Jeff}, [v, [\text{like}, \text{Mike}]]]]]
\end{array}\right\}$$

(33)
$$\left\{\begin{array}{c}
[\text{Tim}, [pres, [\text{Tim}, [v, [\text{like}, \text{-self}]]]]], \\
[\text{Mike}, [pres, [\text{Mike}, [v, [\text{like}, \text{-self}]]]]], \\
[\text{Jeff}, [pres, [\text{Jeff}, [v, [\text{like}, \text{-self}]]]]],
\end{array}\right\}$$

We can characterize these in syntactic terms fairly simply—(32) is the case of maximal instances (external) merge, while (33) is the case of maximal binding—while the **eachselves* case would simply be the union of the two, making it no surprise that **eachselves* is actually pronounced "themselves and each other". At the very least, the union of (32) and (33) does not form a naturally unifiable set such that we would expect it to be reducible to something like (31).

VI

A lurking concern that may have occurred to the reader during the preceding discussion about plural anaphors is that these are not strictly facts about coordination, but also about plurals, and that therefore, our explanation of these facts in the context of coordination should also apply in the context of plurals. We could address this provisionally, by hypothesizing that plurality and conjunction are the same underlying phenomenon (for a similar hypothesis regarding numerals, see Kayne 2019).

[4] Specifically, various members of the Linguistics Department at the University of Toronto in 2022.

While a full spelling-out of this hypothesis is beyond the scope of this paper, we can put forth some arguments toward the plausibility or such a hypothesis.

If number were a φ-feature like gender and person, we would expect it to behave like one, yet this does not seem to be the case. Consider what happens when different specifications of number, gender, and person respectively are combined with each other—we might call the arithmetic of features. For instance, when a 1st person (*e.g.*, *I*) and 2nd person (*e.g.*, *you*) are conjoined, the result is a 1st person (*e.g.*, *we*)—$1^{\text{st}} + 2^{\text{nd}} = 1^{\text{st}}$—and when a 2nd person is combined with a 3rd person, the result is a 2nd person—$2^{\text{nd}} + 3^{\text{rd}} = 2^{\text{nd}}$—and so on. Or when a masculine (*e.g.*, French *il* "he") is combined with a feminine (*e.g.*, French *elle* "she"), the result is masculine (*e.g.*, *ils* "they(Masc)")—$\text{M} + \text{F} = \text{M}$—setting aside cases where gender is neutralized in plurals. Thus we have a general pattern where the specifications of a feature are hierarchically organized—$1^{\text{st}} > 2^{\text{nd}} > 3^{\text{rd}}$, $\text{M} > \text{F}$—and when features are combined the higher specification wins out. Note, that this rule accurately predict what happens when, for example two 2nd persons are conjoined, or when two neuters are conjoined. Since neither outranks the other, the features are unchanged—a 2nd person and a neuter result respectively.

Now, suppose we assume that number behaved the same way, we could discover the hierarchy by combining different numbers and seeing which wins out. In the case of English, a singular and a plural together result in a plural, which suggests the ranking $\text{Pl} > \text{Sg}$. When we combine two singulars, then, we expect another singular, but what we actually get is a plural. Thus, it seems that grammatical number behaves like natural number, and unlike a φ-feature.

There are two possible conclusions from this. Either number is not a φ-feature, or it is simply a different sort of φ-feature from person and gender. Best scientific practice, however, demands that we at least consider the former until it is falsified either empirically or theoretically, since the latter demands the hypothesis of a new sort of entity. That demand, however, will have to be met in another paper.

VII

This paper has had two results one being an explanation of the phenomenon of grammatical coordinations and the other is the synthesis of two distinct hypotheses regarding grammatical theory. The former, being an empirical result, requires little expansion of, or meditation upon—its strengths, weaknesses, and implications can be judge on its further application to data. The latter, though, may offer further theoretical, metatheoretical, or methodological consequences. For instance, a preliminary glance at previous discussion FormSet and Parallel Derivation suggests that they are two contradictory ways of accounting for coordination and adjunction—indeed, this paper originated from a talk in which the superiority of PD over FS was argued—yet we were able to synthesize them by keeping their respective justifications firmly in mind.

FormSet was justified effectively on minimalist grounds—current theory has hypothesized the workspace, and FormSet is necessary to construct workspaces. Indeed, it is virtually inconceivable that our cognition does not have FormSet. The main function of FormSet with respect to grammatical derivation, then, is preperatory—to set the stage for a derivation. Thus, FormSet should operate before MERGE.

Parallel Derivation, on the other hand, was justified largely on empirical grounds—it predicts a number of thorny empirical issues of coordination. Thus, the theoretical mechanism that Milway (2022) introduces in order to acheive it can be dispensed with so long as we can derive the right structures. The task, then, became the merely technical on of generating Parallel Derivation structures from MERGE operating on FormSet sets, without losing generality.

This synthesis, however, serves to underline a problem which has likely existed since the introduction of the workspace and that is the mostly implicit distinction between sets and sets *qua* syntactic objects. The present work hinges on the conclusion that FormSet sets are not Syntactic Objects despite the fact that Syntactic Objects, in the classic minimalist formulation, are nothing but simple binary sets. We have provisionally stipulated here, following Fong and Oishi (2023) and Marcolli, Chomsky, and Berwick (2025), that FormSet sets and merge sets are not the same sort of thing, but such a stipulation should not be made permanent. That is, it behooves us as theorists to propose a principled distinction between the two.

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